

KIANA, AK, USA

WMO: 701196

Lat: 66.976N

Lon: 160.437W

Elev: 51

StdP: 100.71

Time Zone: -9.00 (NAK)

Period: 09-19

WBAN: 00102

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.2	-37.7	-36.9	0.1	-33.5	-35.8	0.1	-32.7	10.3	-6.7	9.0	-7.9	1.0	340	0.759

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	24.1	14.9	22.4	14.4	20.6	13.7	16.1	21.9	15.2	20.4	14.4	19.3	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	9.9	17.2	12.9	9.3	16.5	12.3	8.9	15.8	45.0	21.8	42.6	20.3	40.4	19.3	18.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.5	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.0	-19.7	-15.6	-14.1	-4.4	4.7	12.8	14.2	11.2	5.6	-1.9	-13.0	-16.9		
	DBStd	13.82	11.60	9.42	7.14	6.32	5.45	3.90	3.20	3.34	4.26	4.86	7.80	8.37		
	HDD10.0	5070	921	716	747	431	176	17	4	25	140	370	691	833		
	HDD18.3	7802	1179	949	1006	681	423	169	133	220	382	628	941	1091		
	CDD10.0	319	0	0	0	0	11	102	134	64	9	0	0	0		
	CDD18.3	9	0	0	0	0	0	4	5	0	0	0	0	0		
	CDH23.3	109	0	0	0	0	2	55	51	1	0	0	0	0		
	CDH26.7	16	0	0	0	0	0	11	4	0	0	0	0	0		
(14) Wind	WSAvg	2.7	2.6	3.0	2.5	2.7	2.8	2.9	2.5	2.5	2.6	2.8	2.6	2.6		
(15) Precipitation	PrecAvg	464	33	35	26	23	20	27	57	71	50	50	38	35		
	PrecMax	596	68	107	95	64	52	49	117	173	92	110	99	75		
	PrecMin	347	6	7	8	6	5	13	20	23	22	16	8	18		
	PrecStd	67	17	22	18	14	10	9	21	31	19	22	24	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	2.1	3.0	11.2	22.1	28.3	27.3	23.0	18.0	10.4	3.2	2.1		
		MCWB	0.1	1.7	0.3	5.6	12.3	16.5	16.2	14.6	11.2	3.7	1.5	0.5		
	2%	DB	-0.2	0.3	0.9	8.6	18.0	24.6	24.4	20.7	15.8	7.3	1.3	0.2		
		MCWB	-1.3	-0.5	-1.0	3.8	10.5	14.3	15.5	14.3	10.1	4.1	0.5	-1.1		
	5%	DB	-2.2	-1.3	-1.4	6.3	15.6	22.4	22.7	18.7	13.8	5.9	-0.8	-2.1		
		MCWB	-3.0	-2.1	-2.8	2.4	9.2	13.7	15.2	13.6	9.8	3.5	-1.4	-2.8		
	10%	DB	-4.6	-2.9	-3.7	4.1	13.0	20.7	21.0	17.2	12.2	4.3	-2.8	-5.1		
		MCWB	-5.4	-3.8	-5.2	0.9	8.0	12.9	14.6	12.8	9.1	2.6	-3.5	-5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	1.5	1.1	5.8	12.5	16.7	17.6	16.2	12.4	6.0	1.7	0.7		
		MCDB	1.6	1.8	2.3	10.5	20.8	25.7	25.0	19.9	15.7	6.9	2.8	1.7		
	2%	WB	-1.1	-0.2	-0.8	4.2	10.9	15.4	16.5	15.2	11.5	4.9	0.6	-0.7		
		MCDB	-0.3	0.4	0.6	8.2	16.9	22.0	22.8	18.8	13.9	6.7	1.3	0.1		
	5%	WB	-2.5	-2.1	-2.8	2.7	9.7	14.5	15.7	14.4	10.5	3.9	-1.2	-2.8		
		MCDB	-1.7	-1.4	-1.5	5.9	14.6	20.6	21.2	18.0	12.5	5.5	-0.7	-1.7		
	10%	WB	-4.5	-3.7	-4.9	1.4	8.6	13.7	14.9	13.5	9.6	2.9	-3.5	-5.6		
		MCDB	-3.6	-2.7	-3.6	4.0	12.7	19.4	20.0	16.3	12.0	4.2	-2.8	-5.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.8	10.3	9.8	9.5	9.9	8.3	7.8	7.9	5.6	6.1	6.4		
		MCDBR	7.0	6.4	8.8	10.0	13.1	13.1	12.4	10.5	10.0	5.7	6.0	5.9		
	5% WB	MCWBR	6.4	6.1	7.0	6.2	6.9	5.6	5.3	5.4	5.7	3.7	5.6	5.2		
		MCWBR	7.5	6.7	7.8	9.3	11.7	10.9	11.0	8.6	7.0	4.8	6.0	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.177	0.238	0.256	0.322	0.355	0.349	0.363	0.358	0.316	0.256	0.171	N/A		
	taud		1.904	2.128	2.251	2.118	2.242	2.419	2.399	2.428	2.490	2.318	1.926	N/A		
	Ebn at Noon		196	609	814	823	836	857	829	792	747	594	188	0		
	Edh at Noon		13	49	75	114	114	98	98	86	63	40	11	0		
(44) All-Sky Solar Radiation	RadAvg		0.07	0.58	2.32	4.47	5.42	5.89	4.70	3.46	2.13	0.77	0.15	0.00		
	RadStd		0.01	0.07	0.19	0.28	0.57	0.55	0.41	0.46	0.26	0.09	0.02	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+1.43	N/A	N/A	+1.00	+0.82	N/A	-475	-510	+65	+1

Nomenclature: See separate page